

PROGRAM-5

Demonstrate CI integration by running jenkins in a Docker container

Program Structure:

Dockerfile
docker-compose.yml

Steps 1: Check whether docker service is running

Steps 2: Create a folder prog-5 and navigate to it

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ docker version
Client: Docker Engine - Community
Version: 28.5.1
API version: 1.51
Go version: go1.24.8
Git commit: e180ab8
Built: Wed Oct 8 12:17:26 2025
OS/Arch: linux/amd64
Context: default
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.51/version": dial unix:///var/run/docker.sock: connect: permission denied
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ docker compose version
Docker Compose version v2.40.0-desktop.1
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ mkdir p-5
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ cd p-5
```

Step 3: Create a Dockerfile

```
FROM jenkins/jenkins:lts
#Go to the root user to install tools
USER root
#Install docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io
#Go back to jenkins user
USER jenkins
```

Step 4: Create a docker-compose.yml and add

```

version '3.8'
scripts:
  jenkins:
  build: .
  container_name: jenkins-ci
  ports:
    - "8080:8080"
    - "50000:50000"
  volumes:
    - jenkins_home:/var/jenkins_home
    - /var/run/docker.sock:/var/run/docker.sock
volumes:
  jenkins_home

```

Step 5: Build and run the jenkins

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-5$ sudo docker build -t prog-5 .
[sudo] password for _1RV24MC019_anusha_s:
[+] Building 2.9s (6/6) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 242B                             0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts  2.7s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c378278279c8688efd 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io 0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:0b6f740e3c087354dad0cdc19fd188f0dfa6c4bdfbc1c 0.0s
=> => naming to docker.io/library/prog-5                        0.0s

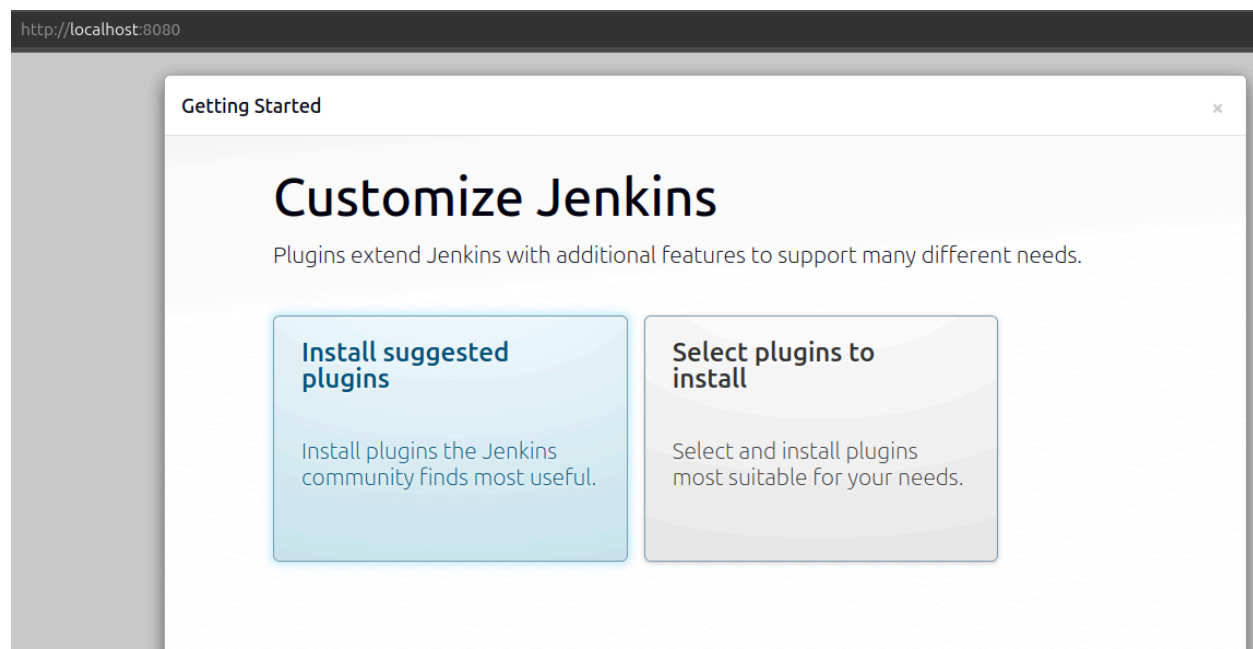
```

```

_L1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-5$ sudo docker run --name jenkins-ci -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home -v /var/run/docker.sock:/var/run/docker.sock prog-5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-11-21 04:24:30.715+0000 [id=1] INFO winstone.Logger#LogInternal: Beginning extraction from war file
2025-11-21 04:24:30.912+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-11-21 04:24:31.063+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea; jvm 21.0.9+10-LTS
2025-11-21 04:24:32.098+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2025-11-21 04:24:32.230+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2025-11-21 04:24:33.076+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2025-11-21 04:24:33.324+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@1c12f3ee[Jenkins v2.528.2/,b=file:///var/jenki
ns_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreNestedHandler@6d467c87{STARTED}]
2025-11-21 04:24:33.346+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@5b3f3ba0(HTTP/1.1, (http/1.1)){0.0.0.0:8080}
2025-11-21 04:24:33.369+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@4a883b15{STARTING}[12.0.25,sto=0] @4117ms
2025-11-21 04:24:33.371+0000 [id=40] INFO winstone.Logger#LogInternal: Winstone Servlet Engine running: controlPort=disabled
2025-11-21 04:24:33.619+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.2
2025-11-21 04:24:33.786+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-11-21 04:24:33.877+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2025-11-21 04:24:35.665+0000 [id=59] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2025-11-21 04:24:35.676+0000 [id=59] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-11-21 04:24:35.677+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-11-21 04:24:36.165+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-11-21 04:24:36.166+0000 [id=46] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-11-21 04:24:36.167+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-11-21 04:24:36.170+0000 [id=54] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-11-21 04:24:36.254+0000 [id=51] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> d96c8ced84bc4646a7861b5f1b612cd7
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>

```

Step 6: Verify whether jenkins is running by navigating to <http://localhost:8080>



Step 7: Display the docker images and remove the prog-5 image by using

sudo docker image

sudo docker remove prog-5

